

Telecom Churn Analysis

Overview

The telecom company is experiencing a significant churn rate of 26.54%, with 1,869 customers churned out of a total customer base of 7,043. The total charges amount to \$16M. The churn rate of **26.54%** is concerning and suggests a need for better customer retention strategies.

Detailed Customer Analysis

Customers by Age Category and Gender

Of the 7043 customers who were grouped into male and female and further into “Under 60” and “Senior” as age groups,

- **Under 60:** Female (41.45%), Male (42.34%)
- **Senior:** Female (8.06%), Male (8.15%)

Implication: The majority of customers are under 60, with seniors forming a smaller proportion. There's a balanced gender distribution.

Recommendations:

- Focus marketing campaigns on younger customers' preferences for flexibility and innovation.
- Design targeted offerings, such as reliability-focused plans, to attract more senior customers.

Customers by Contract Type

- Month-to-month: 55.02%
- One year: 24.07%
- Two year: 20.91%

Implication: A preference for short-term contracts, which could correlate with higher churn due to lower commitment.

Recommendation:

- Offer discounts and benefits (e.g., free upgrades) for one-year and two-year contracts to encourage customers to migrate to long-term commitments.

Customers by Payment Method

- Electronic check: 33.58%
- Mailed check: 22.89%
- Bank transfer (automatic): 21.92%
- Credit card (automatic): 21.61%

Implication: Majorly of the customers (33.58%) prefers electronic check usage which has the ease of cancellation.

Recommendations:

- Maintain diverse payment options to accommodate all customer preferences.
- Simplify and secure payment processes, especially for electronic check users, as they may be more prone to cancellation.

Customers by Partner and Dependents

- Without partner or dependents: 46.57%
- With partner and dependents: 24.83%
- With partner and without dependents: 23.47%
- Without partner and with dependents: 5.13%

Implication: 46.57% of customers do not have partners or dependents, which might be interpreted to low engagement or commitment.

Recommendations:

- Develop promotions targeting families with dependents to increase engagement.
- Consider loyalty programs for non-partnered customers without dependents, as they may be more stable long-term.

Customers by Internet Service

- Fiber optic: 43.96%
- DSL: 34.37%
- No internet service: 21.67%

Implication: Fiber optic is the most popular service, indicating customer preference for high-speed connections.

Recommendations:

- Invest in fiber optic infrastructure to enhance service quality.
- Market DSL as a cost-effective alternative to attract more price-sensitive customers.

Customers by Paperless Billing and Phone Service

- With paperless billing and phone service: 53.73%
- Without paperless billing or phone service: 41.9%
- Without paperless billing and with phone service: 36.59%
- With paperless billing and without phone service: 5.49%

Implication: The majority prefer paperless billing and have phone service. **Recommendation:** Ensure seamless and secure paperless billing processes.

Customers by Monthly Charges

Monthly charges have a peak at lower brackets, and 16% of the customer population is charged between \$18-\$21.60 monthly.

Implication: A concentration of customers with lower monthly charges suggests price sensitivity.

Recommendation: Provide add-ons or upgrades for higher-paying customers to maximize revenue.

Detailed Churn Analysis

Churn by Contract Type

- Month-to-month contracts show the highest churn rate at approximately 88.55%.
- One-year contracts have moderate churn (8.88%).
- Two-year contracts demonstrate the lowest churn (2.57%).

Implication: Month-to-month contracts represent the largest segment at 88.55% of churned customers.

Recommendations:

- Create compelling incentives for month-to-month customers to shift to one or two-year contracts.
- Develop loyalty bonuses that increase with contract length.

- Implement specialized retention programs for new customers in their first 12 months.
- Highlight the value of longer-term contracts through targeted marketing campaigns.
- Educate month-to-month customers on cost savings and benefits of annual contracts.

Churn by Gender and Age Group

Of the 1869 Customers who were grouped into male and female, and further into “Under 60” and “Senior” as age groups. Female and male customers have similar churn rates of 26.92% and 26.16%, respectively. Tailored strategies may be unnecessary, but addressing shared concerns will help overall retention.

Senior customers have a significantly higher churn rate (41.68%) compared to customers under 60 (23.61%).

Implication: There is a higher churn rate among senior customers than among younger customers. This may indicate dissatisfaction with pricing or service expectations.

Recommendations:

- Launch senior-friendly plans focusing on reliability and affordability.
- Provide personalized support to address specific concerns for senior users.
- Use feedback channels to address shared pain points affecting both genders.

Churn by Tenure

New customers (0-10 months) show extremely high churn rates of up to 61.99%. Churn steadily decreases as tenure increases, dropping to 1.66% at 72 months. Clear correlation between longer tenure and higher customer retention.

Implication: Churn is highest for new customers in the first 12 months, suggesting dissatisfaction during onboarding or early service stages.

Recommendations:

- Implement onboarding programs with personalized check-ins during the first three months.
- Implement proactive check-ins and satisfaction surveys at months 3, 6, and 9.
- Offer introductory discounts or free trial features to build loyalty early.
- Highlight the value of your services to improve early-stage satisfaction.

Churn by Monthly Charge

- Higher monthly charges generally correlate with increased churn rates.
- Several charge bands around \$68-105 show particularly high churn rates.

Implication: Customers with higher monthly charges experience greater churn, possibly due to value concerns.

Recommendations:

- Introduce tiered pricing plans to cater to different income levels.
- Reward high-paying customers with loyalty perks or free add-ons.

Churn by Payment Method

Electronic check has the highest churn at 57.3%, followed by mailed check at 16.48% compared to bank transfer (automatic) and credit card (automatic) payments at 26.21% combined.

Implication: Electronic check has the highest churn at 57.3%, possibly due to the ease of cancellation.

Recommendations:

- Promote automatic bank transfers and credit card payments through discounts or rewards programs.
- Review the electronic check payment experience to identify potential issues.
- Improve the ease and security of electronic check payments.

Churn by Internet Service

- Fiber optic users show higher churn (24.56%) than DSL users (24.56%).
- Customers without internet service have the lowest churn (6.05%).

Implication: Fiber optic has the highest churn at 69.4%, indicating dissatisfaction or unmet expectations.

Recommendations:

- Invest in infrastructure improvements, especially fiber optic internet, to meet customer expectations.
- Communicate transparently about service updates or interruptions.
- Enhance fiber optic service quality or provide cost-effective bundles to retain customers.
- Conduct surveys to identify common complaints.

Churn by Dependents and Partner

Customers without partners and dependents show the highest churn (60.09%). Next are customers with partners but without dependents, at 22.47%. Having dependents appears to reduce churn significantly across categories.

Implication: High churn is predominant amongst customers without partners and dependents (60.09%) and customers with partners but without dependents (22.47%).

Recommendations:

- Create family-oriented plans or discounts for customers with dependents.
- Build loyalty by offering partner-oriented incentives.
- Introduce family plans that incentivize adding dependents to existing accounts.

Churn by Phone service and paperless billing

Phone service and paperless billing show notable patterns:

- Customers with phone service and paperless billing have the highest churn (68.59%).
- Customers without phone service but with paperless billing have 6.31% churn.
- Customers with phone service but without paperless billing have 22.31% churn.
- Lowest churn (2.78%) is among customers without paperless billing and phone service.

Implication: Customers with phone service and paperless billing have the highest churn at 68.59%, which could be due to dissatisfaction.

Recommendations:

- Investigate and resolve issues that might cause dissatisfaction with these services.
- Offer promotions for customers opting into bundled services combining phone and internet.
- Review the paperless billing experience to address potential customer concerns.

Conclusion

The telecom company's 26.54% churn rate represents a significant opportunity for improvement. By targeting the high-risk segments identified and implementing the recommended strategies, the company could potentially reduce churn by 30-40%, resulting in substantial revenue retention and increased customer lifetime value.